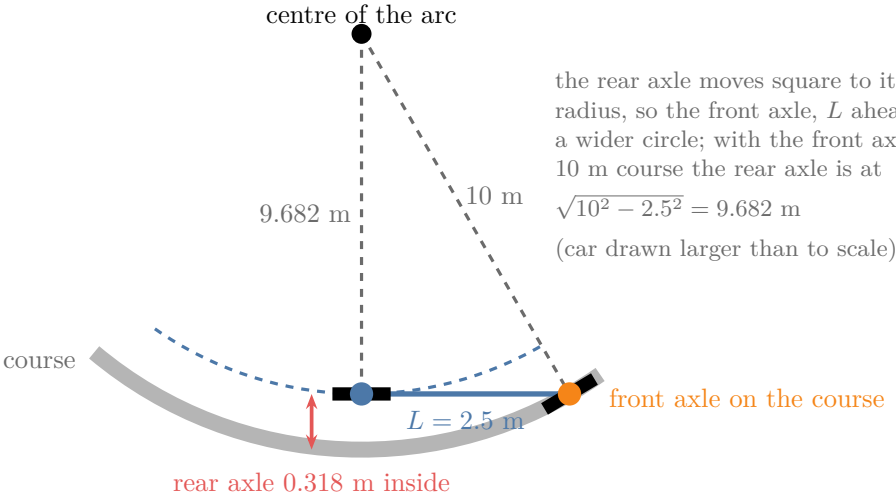


the rear axle moves square to its own radius, so the front axle, L ahead, sits on a wider circle; with the front axle on the 10 m course the rear axle is at

$\sqrt{10^2 - 2.5^2} = 9.682$ m

(car drawn larger than to scale)